Review 2 Commitment

Form BM04. Completed within 3 working days after Review 2 and confirmed by both reviewers and the supervisor. This document is the sole basis for Defense 1.1 and the Viva. One block per main workflow (section 6) and one block per member (section 7): duplicate or delete blocks to match the team. Export to PDF. File name: SEP490_<ProjectCode>_R2_Commitment.pdf

1. Project Information

Project code	FA26SE249
Project name	PrintGrid: Development of a Distributed 3D Printing Fulfillment and Scheduling Platform for a Network of Independent Printing Labs
Team	Nguyen Xuan Tan (SE181599, Leader) Ã‚Â· Dau Nguyen Bao Tuan (SE185029) Ã‚Â· Nguyen Quang Minh (SE181684) Ã‚Â· Truong Quang Nhat (SE181682)
Supervisor	Nguyen Tan Phuc (phucnt40@fpt.edu.vn)
Review 2 date	02/11/2026
Reviewer 1	Reviewer 1
Reviewer 2	Reviewer 2
Version	1.0
Date	02/11/2026

2. Final Registration Form

Final version	R2-Nov09
Approved at Review 2 (Yes/No)	Yes
Attached file name	SEP490_FA26SE249_R2_Registration.pdf

After this point the project scope does not change.

3. Mandatory Features

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-01	Customer registration and authentication	Customer	Session on register, login and refresh; duplicate email 409; wrong password and unknown email 401; cross-user 404.	Tuan	
F-02	Model upload, evidence record and 3D preview	Customer	SHA-256 stored; receipt email <=1 min; STL/3MF preview; invalid file rejected with reason.	Tuan	
F-03	Quote draft and staff publication	Customer; Ops Manager	Draft p95 <=60 s; out-of-band edit rejected; publish audit; frozen version reproduces price.	Tuan	
F-04	Order placement and staged deposit	Customer; Ops Manager	PaymentPending until deposit %; no production before Confirmed; unpaid balance blocks packing; idempotent.	Tuan	
F-05	Lab registration, machines and inventory ledger	Lab Manager	Second lab per owner 409; approval needs machine and material; stock	Nhat	

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
			code and running total; 1-5% tolerance.		
F-06	Offer acceptance, production and lab self-QC	Lab Manager; Lab Operator	Accept only in 2 h window with matching hash; completion needs actuals and photo; rejected proof returns to InProgress.	Minh; Nhat	

4. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
NFR-PERF-001	Response time	Draft quote p95 <=60 s (preliminary <=5 s)	Timestamp 100 runs in k6	Minh	
NFR-PERF-002	Response time	Reschedule wall-clock <=30 s (mean <=15 s)	Simulator, 50 concurrent events	Minh	
NFR-SEC-007	Security	Unauthorized file access = 0; 100% logged	Cross-lab and expiry tests; audit sample	Tuan	

5. Out of Scope

Item	Reason
Carrier API integration and live tracking	Manual waybill only; no carrier contract
Resin and SLS printing	MVP restricted to FDM so slicing, estimation and QC stay verifiable
Native mobile apps, lab e-wallets, automated payouts, in-platform 3D modelling	Resource limit; payouts are computed in-system, settled by runbook

6. Workflow Commitments

Minimum number of main workflows = number of team members. Each workflow has exactly one owner. Each mandatory feature belongs to at least one workflow.

6.1 Workflow WF-01

Workflow name	Customer order to quote and deposit
Owner (one member)	Dau Nguyen Bao Tuan
Supporting members	Nguyen Xuan Tan
Actors	Customer; Ops Manager; Engine
Trigger	Customer uploads and configures
Business outcome	Committed price and date plus paid deposit create a Confirmed order
Main entity state path	Quote Pending->Draft->Ready->Converted; Order PaymentPending->Confirmed
Activity diagram (Report 4 section)	Report 4 Ã‚Â§3.1

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-01	Customer registration and authentication	Session on register, login and refresh; duplicate email 409; wrong password and unknown email 401; cross-user 404.	TC-F01-01..03
F-02	Model upload, evidence record and 3D preview	SHA-256 stored; receipt email <=1 min; STL/3MF preview; invalid file rejected with reason.	TC-F02-01..03
F-04	Order placement and staged deposit	PaymentPending until deposit %; no production before Confirmed; unpaid balance blocks packing; idempotent.	TC-F04-01..04

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	*Main*	Seeded customer, model and pricing version; deposit 50%	Login -> upload -> configure -> quote -> Ops publish -> deposit -> Confirmed	Quote Ready; order Confirmed; deposit once	
DS-02	*Exception*	Published quote past expiry	Attempt checkout	Checkout rejected; re-quote required; no order	

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (â‰¥ 3 per feature)	12	Yes	Dau Nguyen Bao Tuan
End-to-end workflow test (â‰¥ 1)	1	Yes	Dau Nguyen Bao Tuan
State transition tests (â‰¥ 1 per transition)	2	Yes	Dau Nguyen Bao Tuan

Test type	Count	Automated in CI	Responsible member(s)
Total for this workflow	15		

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	2/2
Test pass rate for this workflow	>=90%
Open Critical / High defects	*0*

Reviewer note

Note	

6.2 Workflow WF-02

Workflow name	Placement, offer and lab acceptance
Owner (one member)	Nguyen Quang Minh
Supporting members	Nguyen Xuan Tan
Actors	Engine; Lab Manager; Ops Manager
Trigger	Order Confirmed or placement requested
Business outcome	Capable lab accepts; job scheduled without double booking
Main entity state path	Job Pending->Assigned->Accepted
Activity diagram (Report 4 section)	Report 4 Ã‚Â§3.2

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-06	Offer acceptance, production and lab self-QC	Accept only in 2 h window with matching hash; completion needs actuals and photo; rejected proof returns to InProgress.	TC-F06-01..04
F-07	Backward scheduling and event-driven rescheduling	DB rejects overlap; decline/failure repair <=30 s; quantity split before date change; proposal expires at 48 h.	TC-F07-01..04

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-03	*Main*	Confirmed order; three capable labs	Rank labs -> accept best offer with hash acknowledgement	Job Assigned->Accepted; slot reserved; no overlap	
DS-04	*Exception*	Top lab declines or 2 h timeout	Decline; engine re-ranks	Lab excluded; job Reassigned; decision log records reason	

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (â‰¥ 3 per feature)	6	Yes	Nguyen Quang Minh
End-to-end workflow test (â‰¥ 1)	1	Yes	Nguyen Quang Minh
State transition tests (â‰¥ 1 per transition)	2	Yes	Nguyen Quang Minh
Total for this workflow	9		

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	2/2
Test pass rate for this workflow	>=90%
Open Critical / High defects	*0*

Reviewer note

Note	

6.3 Workflow WF-03

Workflow name	Production, self-QC and lab-to-hub batch
Owner (one member)	Truong Quang Nhat
Supporting members	Nguyen Quang Minh
Actors	Lab Operator; Ops Manager; Hub QC
Trigger	Lab accepts the job
Business outcome	Proof-approved jobs leave in a sealed batch
Main entity state path	Job Accepted->InProgress->AwaitingInspection; QcProof None->Pending->Approved; Batch Sealed
Activity diagram (Report 4 section)	Report 4 Ã‚Â§3.3

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-06	Offer acceptance, production and lab self-QC	Accept only in 2 h window with matching hash; completion needs actuals and photo; rejected proof returns to InProgress.	TC-F06-01..04
F-08	Hub receipt, inspection, reprint and shipment	Batch receipt reconciles jobs; checklist by grade; reprint after attribution; same-customer merge; waybill recorded.	TC-F08-01..04

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-05	*Main*	Accepted job at lab	Start -> complete with actuals and photo -> Ops	Job AwaitingInspection-	

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
			approves -> seal -> hub receives	>InTransit->Completed; batch Received	
DS-06	*Exception*	Missing or blurred proof	Ops rejects proof	Job back to InProgress; batch cannot be sealed	

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (â‰¥ 3 per feature)	6	Yes	Truong Quang Nhat
End-to-end workflow test (â‰¥ 1)	1	Yes	Truong Quang Nhat
State transition tests (â‰¥ 1 per transition)	2	Yes	Truong Quang Nhat
Total for this workflow	9		

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	2/2
Test pass rate for this workflow	>=90%
Open Critical / High defects	*0*

Reviewer note

Note	

6.4 Workflow WF-04

Workflow name	Hub inspection, reprint and delivery
Owner (one member)	Truong Quang Nhat
Supporting members	Dau Nguyen Bao Tuan
Actors	Hub QC; Hub Fulfillment; Customer
Trigger	Batch received at hub
Business outcome	Inspected, consolidated and delivered order, or attributed reprint
Main entity state path	Job AwaitingInspection->Completed/Failed->Reassigned; Order QualityCheck->Shipping->Delivered
Activity diagram (Report 4 section)	Report 4 Ã‚Â§3.4

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-08	Hub receipt, inspection, reprint and shipment	Batch receipt reconciles jobs; checklist by grade; reprint after attribution; same-customer merge; waybill recorded.	TC-F08-01..04

Feature ID	Feature	Acceptance criteria	Test case IDs

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-07	*Main*	Received batch passes inspection	Inspect -> consolidate -> waybill -> deliver	Order Shipping->Delivered; one waybill	
DS-08	*Exception*	Batch fails inspection, lab fault	Record defect and attribution -> reprint	Reprint created; cost bearer Lab; original Failed	

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (â‰¥ 3 per feature)	3	Yes	Truong Quang Nhat
End-to-end workflow test (â‰¥ 1)	1	Yes	Truong Quang Nhat
State transition tests (â‰¥ 1 per transition)	2	Yes	Truong Quang Nhat
Total for this workflow	6		

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	2/2
Test pass rate for this workflow	>=90%
Open Critical / High defects	*0*

Reviewer note

Note	

6.5 Workflow WF-05

Workflow name	Ops exception and date-change decision
Owner (one member)	Nguyen Xuan Tan
Supporting members	Nguyen Quang Minh
Actors	Ops Manager; Customer; Engine
Trigger	Forecast lateness or no feasible placement
Business outcome	Re-planned schedule or approved or expired date change
Main entity state path	DateChangeRequest Proposed->Approved/Rejected/Expired
Activity diagram (Report 4 section)	Report 4 Ã‚Â§3.5

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-07	Backward scheduling and event-driven rescheduling	DB rejects overlap; decline/failure repair <=30 s; quantity split before date change; proposal expires at 48 h.	TC-F07-01..04
F-09	Operations monitoring, override and SLA	At-risk list by slack; override needs reason; SLA export matches history; suspension blocks new work.	TC-F09-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-09	*Main*	Job forecast late; customer contactable	Ops proposes date -> customer approves	Request Approved; schedule re-planned; customer notified	
DS-10	*Exception*	No customer answer within 48 h	Let request expire	Status Expired; escalation created; no silent change	

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (â‰¥ 3 per feature)	6	Yes	Nguyen Xuan Tan
End-to-end workflow test (â‰¥ 1)	1	Yes	Nguyen Xuan Tan
State transition tests (â‰¥ 1 per transition)	2	Yes	Nguyen Xuan Tan
Total for this workflow	9		

Pass criteria at Defense 1.1

Criterion	Committed
Runs end-to-end on staging with seeded data	*Yes*
Main scenario passes	*Yes*
Exception scenarios passing	2/2
Test pass rate for this workflow	>=90%
Open Critical / High defects	*0*

Reviewer note

Note	

7. Individual Commitments

One block per member. Every member owns at least one main workflow and 2â€“3 parts. Each test case and each fix is counted for one member only.

7.1 Member 1

Full name	Nguyen Xuan Tan (SE181599)
Student ID	SE181599
Role in team	Team leader; PM/BA; backend

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-05	
Mandatory features owned	F-03, F-04, F-09	
Modules / entities owned	Customer; Admin/audit; integration	
Design decisions owned (D-IDs)	D1, D3	
NFRs responsible	NFR-SEC-007, NFR-SEC-008	

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	18	15/11/2026
Of which automated in CI	10	
Demo scenarios presented	DS-09, DS-10	
Required fixes owned (section 9)	G-01, G-02, G-03, SR-01, SR-08, SR-09, SR-11, SR-15	
Documentation sections (Report / section)	BM01, BM02, BM04; Report 4 Ã‚Â§1, Ã‚Â§3.5	

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
â˜ Verified *â˜ Partly verified* *â˜ Not verified*		

Member confirmation

Signature / date	

7.2 Member 2

Full name	Dau Nguyen Bao Tuan (SE185029)
Student ID	SE185029
Role in team	Frontend engineer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-01	
Mandatory features owned	F-01, F-02, F-03, F-04	
Modules / entities owned	Customer portal; Lab/Hub UI; design system	
Design decisions owned (D-IDs)	D5	
NFRs responsible	NFR-PERF-001	

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	15	15/11/2026
Of which automated in CI	8	
Demo scenarios presented	DS-01, DS-02	
Required fixes owned (section 9)	G-06, SR-07	
Documentation sections (Report / section)	BM01, BM04; Report 4 Ã‚Â§5 UI/UX	

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
â˜ Verified *â˜ Partly verified* *â˜ Not verified*		

Member confirmation

Signature / date	

7.3 Member 3

Full name	Nguyen Quang Minh (SE181684)
Student ID	SE181684
Role in team	Algorithm engineer

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-02	
Mandatory features owned	F-06, F-07	
Modules / entities owned	Scheduling engine; simulator; calibration	
Design decisions owned (D-IDs)	D2	
NFRs responsible	NFR-REL-004, NFR-PERF-002	

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	14	15/11/2026
Of which automated in CI	9	
Demo scenarios presented	DS-03, DS-04	

Commitment	Committed (count / IDs)	Due
Required fixes owned (section 9)	G-04, G-05, SR-04, SR-05, SR-06, SR-10, SR-13, SR-14	
Documentation sections (Report / section)	BM01, BM04; Report 4 Ã‚Â§2, Ã‚Â§3.2, Ã‚Â§3.5	

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
â˜ Verified *â˜ Partly verified* *â˜ Not verified*		

Member confirmation

Signature / date	

7.4 Member 4

Full name	Truong Quang Nhat (SE181682)
Student ID	SE181682
Role in team	Backend/QA; Lab and Hub

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-03, WF-04	
Mandatory features owned	F-05, F-06, F-08	
Modules / entities owned	Lab; Hub; QC and shipment	
Design decisions owned (D-IDs)	D4	
NFRs responsible	NFR-ACC-001, NFR-SCALE-001	

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	12	15/11/2026
Of which automated in CI	5	
Demo scenarios presented	DS-05, DS-06, DS-07, DS-08	
Required fixes owned (section 9)		
Documentation sections (Report / section)	BM01, BM04; Report 4 Ã‚Â§3.3, Ã‚Â§3.4	

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
â˜ Verified *â˜ Partly verified* *â˜ Not verified*		

Member confirmation

Signature / date	

7.5 Member 5

Full name	
Student ID	
Role in team	

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned		
Mandatory features owned		
Modules / entities owned		
Design decisions owned (D-IDs)		
NFRs responsible		

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written		
Of which automated in CI		
Demo scenarios presented		
Required fixes owned (section 9)		
Documentation sections (Report / section)		

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
â˜ Verified *â˜ Partly verified* *â˜ Not verified*		

Member confirmation

Signature / date	

8. Summary and Cross-check

8.1 By workflow

Workflow	Owner	Features	Demo scenarios (main / exception)	Test cases	Automated
WF-01	Tuan	F-01..F-04	2 / 0	15	8
WF-02	Minh	F-06, F-07	2 / 0	14	9
WF-03	Nhat	F-06, F-08	2 / 0	12	5
WF-04	Nhat	F-08	2 / 0	8	3
WF-05	Tan	F-07, F-09	2 / 0	10	7
Total				59	32

8.2 By member

Member	Workflows owned	Features	Test cases	Automated	Demo scenarios	Review 2 verification
Tan	WF-05	F-03, F-04, F-09	18	10	DS-09, DS-10	Verified
Tuan	WF-01	F-01..F-04	15	8	DS-01, DS-02	Verified
Minh	WF-02	F-06, F-07	14	9	DS-03, DS-04	Verified
Nhat	WF-03, WF-04	F-05, F-06, F-08	12	5	DS-05..DS-08	Verified
Member 5 (not applicable)	-	0	0	0	-	N/A
Total						

The totals in 8.1 and 8.2 must both equal the committed numbers in 8.3.

8.3 Committed Numbers

Number of main workflows	5
Number of demo scenarios (main / exception)	10 main / 5 exception
Committed functional test cases	59
Of which automated in CI	32 (54%)

8.4 Eligibility Thresholds for the Viva

Criterion	Minimum threshold	Agreed threshold
Main workflows	*100% run end-to-end*	5 of 5
Demo scenarios	*100% main; â‰¥ 80% exception*	10 / 5 (>=80%)
Test case count	*â‰¥ committed number*	59
Test pass rate	*â‰¥ 90% overall; 100% main-path tests of mandatory features*	>=90%
Automation	*â‰¥ 50% run in CI*	>=50%

Criterion	Minimum threshold	Agreed threshold
Defects	*No open Critical or High defects*	0
Individual	*Each member demos the workflow they own and meets their delivery commitments*	4 of 4

9. Required Fixes

Items from the Review 2 minutes and Planned items from Form BM02.

ID	Item	Related workflow	Owner	Due
RF-01	Public sample/pricing/network pages lack approved FR	WF-01	Tan	08/11/2026
RF-02	Add SampleModel to ERD and DB	WF-01	Tan	08/11/2026
RF-03	Gateway webhooks and idempotency for payments	WF-01	Tan	20/11/2026
RF-04	Top-k offer fan-out and first-accept	WF-02	Minh	20/11/2026
RF-05	Shipping quote line and delivery cost	WF-02	Minh	15/11/2026

10. Review 2 Conclusion (completed by reviewers)

â˜ Pass â˜ Conditional pass â˜ Re-review

11. Confirmation

Role	Full name	Signature	Date
Reviewer 1			
Reviewer 2			
Supervisor			
Team leader			